

Maria Fay Garcia

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riddlework.github.io

EDUCATION

University of Arizona

Tucson, AZ

College of Science, W.A. Franke Honors College

Bachelor of Science with Honors in Computer Science and Mathematics

September 2021 – August 2025

GPA: 3.48

EXPERIENCE

Technical Research Project – Honors Thesis

January 2025 – August 2025

University of Arizona

Tucson, Arizona

- Conducted independent research on virtual memory, process isolation, trusted execution boundaries, and complex technical system design for an honors thesis in computer science.
- Designed, implemented, and documented a C prototype exploring an alternative architecture for shared library execution.
- Documented system design considerations, implementation results, and open research questions in a long-form thesis report.
- Managed a self-directed technical project from proposal through final submission, incorporating implementation, debugging, documentation, project organization, and advisor feedback.

Research Intern

June 2025

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Bay Area, CA

- Conducted research on foundational theoretical questions of computation, formal systems, and mathematical reasoning through a metamathematical lens.
- Synthesized literature across computability theory, proof theory, and mathematical logic to identify open research directions.

Stanford LINXS Computer Science Summer Research Internship

June 2024 – August 2024

Stanford University

Stanford, CA

- Collaborated with Professor Sara Achour and graduate researchers to prototype an online branch predictor using hyperdimensional computing.
- Contributed to research discussions, technical planning, rapid iterative Python prototype development, documentation, and analysis in a collaborative lab environment.
- Presented research progress, incorporated feedback, and documented findings for an academic research audience.

Undergraduate Research

January 2023 – May 2024

University of Arizona

Tucson, Arizona

- Worked closely with Professor Todd Proebsting on research in combinatorial optimization.
- Developed an interactive web tool to demonstrate theoretical concepts, strengthening experience with user-facing technical design, lightweight web development, and software implementation.

Undergraduate Teaching Assistant

August 2023 – April 2025

University of Arizona

Tucson, Arizona

- Held weekly office hours and supported students one-on-one with patience and clarity to help them achieve a greater understanding of course material.
- Developed rubrics to provide consistent, structured feedback on assignments.
- Coordinated with the teaching team to support course organization, grading workflows, and student communication.

SKILLS

- **Software Development:** prototype development, Python scripting, internal tool development, systems research/programming, data workflow automation, project organization, technical writing, documentation
- **Technical:** Python, SQL, C, Java, JavaScript, React, Flask, LaTeX, Git; Linux/CLI, Docker, AWS, REST APIs, data pipelines, RAG/LLM workflows, debugging, testing, and documentation
- **Communication:** collaborative technical communication, technical documentation, structured feedback, one-on-one support, user-facing technical explanation
- **Languages:** English and Spanish, native proficiency; French, conversational proficiency